TEXORA · LECTURE SHEET

Recipe Prediction for Chemical Finishing

A finish recipe is predicted backwards from a performance specification: the add-on on the fabric is fixed by the test the fabric must pass, and the pad-bath concentration follows from the pick-up. This sheet works that chain in both directions.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 4 · SPECIALIZATION — WET PROCESSING

1. The specification comes first

Colour matching predicts a recipe from a measured colour; finishing predicts a recipe from a **measured performance**: wrinkle-recovery angle, spray rating, flame-test pass, handle grade, dimensional change. The laboratory establishes, for the fabric in question, the relation between **add-on** (grams of finish solids per kilogram of fabric, or % owf) and performance, by padding a series at increasing concentrations, curing, and testing. Prediction then reads the required add-on off that calibration — exactly the build-up logic of Topic 1, with a test result in place of K/S.

Two quantities govern every finishing calculation:

$$\text{Add-on (g/kg)} = \text{bath conc. (g/L)} \times \text{wet pick-up (L/kg)} \quad | \quad \text{Bath conc.} = \text{target add-on} / \text{pick-up}$$

and, for a production run, the liquor actually consumed — which is what purchasing and costing need:

$$\text{Liquor consumed (L)} = \text{fabric mass (kg)} \times \text{pick-up (L/kg)} \quad | \quad \text{Chemical consumed (kg)} = \text{conc. (g/L)} \times \text{liquor (L)} / 1000$$

2. Fully worked prediction — easy-care (crease-resistant) finish

WORKED EXAMPLE 1 — DMDHEU RESIN FINISH, PRODUCTION RUN

Specification: crease-resistant finish on cotton; laboratory calibration shows the target wrinkle recovery is reached at a resin add-on of about 42 g/kg of fabric.

Pad bath: DMDHEU resin 60 g/L; $MgCl_2$ catalyst 12 g/L (20% on resin weight, the usual ratio); softener 20 g/L; wetting agent 2 g/L. Wet pick-up 70% → 0.70 L/kg.

Predicted add-on = 60 × 0.70 = **42 g/kg = 4.2% owf ✓** — the bath was chosen to hit the calibrated add-on at the machine's pick-up.

Production run: 5,000 m of fabric, 1.5 m wide, 220 g/m² → fabric mass = 5,000 × 1.5 × 0.220 = **1,650 kg**.

Liquor consumed = 1,650 × 0.70 = **1,155 L**.

Chemical consumption: resin = 60 × 1,155/1000 = **69.3 kg**; catalyst = 12 × 1,155/1000 = **13.9 kg**; softener = 20 × 1,155/1000 = **23.1 kg**; wetting agent = 2 × 1,155/1000 = 2.3 kg.

In practice the bath is charged with an initial trough volume plus a feed tank and topped up at the same concentration; consumption — not the initial charge — is what the run truly uses, and it is the figure costing must use.

3. Predicting for a fixed add-on — flame-retardant finish

WORKED EXAMPLE 2 — CONCENTRATION FROM A SPECIFIED ADD-ON

A durable flame-retardant specification requires 15% owf of finish solids on the fabric = 150 g/kg. The padder gives 80% pick-up (0.80 L/kg) on this fabric.

Required bath concentration = 150 / 0.80 = **187.5 g/L**.

If the same fabric is padded on a machine giving only 60% pick-up, the bath must rise to 150/0.60 = 250 g/L — or the specification silently fails. Pick-up is therefore verified on the day, on the fabric, before the bath is made.

4. Components of a finishing recipe and their roles

COMPONENT	ROLE AND DOSING NOTE
Active agent	Resin, repellent polymer, softener, antimicrobial — dosed to the calibrated add-on; quoted as supplied product unless stated as solids.
Catalyst	For crosslinking resins, typically 15–25% of the resin weight ($MgCl_2$ for DMDHEU); excess catalyst tenderises cellulose at cure.
Softener / lubricant	Restores handle lost to resin; silicone and polyethylene types; watch for effects on repellency and sewability.
Wetting agent	Ensures rapid, even pick-up, especially on fabrics carrying residual size or softener from earlier processes.
pH control	Bath pH set for stability in the trough and reaction on curing (mildly acidic for resin systems); an unstable bath gels or precipitates and ruins prediction.

5. Where finishing predictions go wrong

- **Pick-up drift:** nip pressure, fabric moisture before padding and bath temperature all move pick-up; a recipe predicted at 70% fails at 60%. Measure pick-up at the start of every run: weigh dry, pad, weigh wet.
- **Bath exhaustion in the trough:** fabric preferentially absorbs some components, so a standing bath changes composition; use a small trough with continuous feed at make-up concentration.
- **Cure variation:** add-on is only potential performance until cured; under-cure wastes chemical, over-cure costs strength. Prediction assumes the calibrated cure schedule.
- **Product strength:** commercial products are supplied at different solids contents; recipes must state product or solids basis explicitly, exactly as dye strength is stated in dyeing.

6. Glossary — key terms

- **Add-on:** finish solids deposited per unit mass of fabric, g/kg or % owf.
- **DMDHEU:** dimethylol dihydroxyethylene urea — the standard durable-press crosslinking resin for cellulose.
- **Cure:** the heat treatment that fixes the finish (crosslinking, polymerisation) after drying.
- **Tendering:** strength loss of cellulose from acid/catalyst action at high temperature.

7. Key points

- Finishing prediction = calibrated add-on → bath concentration via pick-up → consumption via fabric mass.
- Catalyst is dosed on the resin, softener and wetting agent on the bath; quote every component on its stated basis.
- Verify pick-up and cure on the day; the prediction is only as good as those two measurements.

8. Summary

The chain worked in this sheet — specification to add-on (42 g/kg), add-on to bath (60 g/L at 70% pick-up), bath to consumption (69.3 kg resin for a 1,650 kg run) — is the finishing counterpart of dye recipe prediction. Both disciplines share one moral: the recipe is a prediction, and it is the measured variables around the recipe — pick-up, cure, liquor ratio — that decide whether production agrees with it.

9. Practice

1. A water-repellent finish needs an add-on of 24 g/kg. Pick-up is 65%. What bath concentration is required, and how much product is consumed by a 900 kg run?
2. A resin bath of 80 g/L runs at 75% pick-up. What add-on results? If the specification is 66 g/kg, does it pass?
3. Why is catalyst quoted as a percentage of resin rather than in g/L of bath?
4. State two reasons a finish that performed in the laboratory padder fails on the production range at the same bath recipe.

Answer key

1. Conc. = 24/0.65 = 36.9 g/L. Liquor consumed = 900 × 0.65 = 585 L → product = 36.9 × 585/1000 = 21.6 kg.
2. Add-on = 80 × 0.75 = 60 g/kg. It fails the 66 g/kg specification — raise the bath to 66/0.75 = 88 g/L or increase pick-up.
3. The catalyst's job is to crosslink a proportion of the resin actually deposited; its required amount scales with resin add-on, not with bath volume. Quoting it on resin keeps the ratio correct when bath concentration or pick-up changes.
4. Any two of: pick-up differs between laboratory and production padders; cure temperature/time profile differs across the production stenter; trough composition drifts by preferential absorption over a long run; incoming fabric moisture or residual chemicals differ.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib